

## Workshop Practice Lab:

In a Workshop Practice Lab, students learn the basics of engineering and manufacturing practices through hands-on activities. The following are the key points that one should know about Workshop Practice Lab:

1. **Introduction to Workshop Practice:** The Workshop Practice Lab introduces students to various tools and equipment used in a workshop, such as hand tools, power tools, and measuring instruments. Students learn how to use these tools safely and effectively.
2. **Basic Manufacturing Processes:** The Workshop Practice Lab teaches students about basic manufacturing processes such as casting, forging, welding, and machining. Students learn the principles behind each process and how to perform them correctly.
3. **Material Science:** The Workshop Practice Lab teaches students about different materials used in manufacturing, such as metals, plastics, and composites. Students learn about the properties of each material and how to select the right material for a specific application.
4. **Workshop Safety:** Safety is a critical aspect of Workshop Practice. Students learn about workshop safety rules and regulations, such as wearing personal protective equipment (PPE), maintaining a clean and organized workspace, and identifying potential hazards in the workshop.
5. **Measurement and Inspection:** The Workshop Practice Lab teaches students how to use measuring instruments such as calipers, micrometers, and gauges to measure and inspect components accurately. Students learn about different types of measurements, such as linear, angular, and surface measurements.
6. **Prototyping and Fabrication:** The Workshop Practice Lab enables students to design and build prototypes of their own products. They learn about the design process, material selection, and fabrication techniques required to bring their ideas to life.
7. **Teamwork and Communication:** The Workshop Practice Lab encourages students to work in teams to complete various projects. They learn how to communicate effectively, delegate tasks, and work collaboratively to achieve a common goal.

In conclusion, the Workshop Practice Lab provides students with a practical and hands-on learning experience that helps them develop skills in engineering and manufacturing. The lab teaches them about various manufacturing processes, materials, and measurement techniques, as well as workshop safety and teamwork skills.

## **S. No. Mechanical Workshop Duration**

Mechanical workshops are essential for students pursuing mechanical engineering as it provides hands-on experience and practical knowledge. Here are some points to keep in mind regarding the duration of mechanical workshops:

1. The duration of mechanical workshops varies depending on the level of education and the institution. It can range from a few hours to several weeks.
2. In some institutions, mechanical workshops are included as a part of the regular curriculum and may last for a semester or an entire academic year.
3. The duration of mechanical workshops also depends on the type of workshop. For example, a welding workshop may take a shorter duration than a workshop on engine assembly.
4. The aim of the mechanical workshop is to provide students with practical knowledge and experience in the field of mechanical engineering. Therefore, the duration should be long enough for the students to grasp the concepts and apply them practically.
5. Mechanical workshops may also have different levels of complexity, and the duration may vary accordingly. For example, basic workshops may have a shorter duration than advanced workshops.
6. The duration of the workshop may also depend on the availability of resources and equipment in the institution.
7. The duration of the workshop should be planned in a way that allows students to complete the necessary tasks and projects within the given time frame.
8. The duration of the workshop should also include time for students to learn basic safety measures and precautions to be taken while working in a mechanical workshop.

In conclusion, the duration of a mechanical workshop varies depending on several factors such as the level of education, the institution, the type of workshop, and the complexity of the workshop. It is essential to plan the duration in a way that provides students with practical experience and knowledge while ensuring their safety.

### **1 Introduction to Mechanical Workshop Material, Tools and Machines 3 Hrs**

In this section, we will introduce you to the basic materials, tools, and machines commonly used in mechanical workshops. The knowledge gained from this section will provide you with a foundation for understanding and working in a mechanical workshop environment.

1.1 Materials - Materials used in mechanical workshops include metals, plastics, and composites. - Metals are the most commonly used material and can be categorized into ferrous and non-ferrous metals. - Ferrous metals include iron, steel, and cast iron, while non-ferrous metals include aluminum, copper, and brass. - Plastics and composites are also used in some applications due to their lightweight and corrosion-resistant properties.

1.2 Tools - Hand tools are essential in mechanical workshops and include tools such as hammers, pliers, wrenches, and screwdrivers. - Power tools are also used to increase efficiency and include tools such as drills, grinders, and saws. - Measuring tools such as micrometers, calipers, and gauges are used to ensure precision in machining and fabrication. - Safety equipment such as gloves, goggles, and ear protection should always be worn when using tools.

1.3 Machines - Machines used in mechanical workshops include lathes, milling machines, and drill presses. - Lathes are used to turn cylindrical parts, while milling machines are used to cut complex shapes and profiles. - Drill presses are used to accurately drill holes in materials. - Other machines such as welding machines and plasma cutters are used for welding and cutting metals.

In conclusion, understanding the materials, tools, and machines commonly used in mechanical workshops is essential for anyone interested in working in this environment. Make sure to familiarize yourself with the basic concepts and safety procedures before working with any of these tools or machines.

**To study layout, safety measures and different engineering materials (mild steel, medium carbon steel, high carbon steel, high speed steel and cast iron etc) used in workshop.**

When it comes to working in a workshop, there are certain things that you need to keep in mind in order to ensure your safety and the safety of those around you. Additionally, it is important to have a good understanding of the different engineering materials that are commonly used in the workshop. Here are some key points to keep in mind:

### **Workshop Layout and Safety Measures:**

- A well-organized and clean workshop environment is crucial for safety and efficiency.
- Always wear appropriate personal protective equipment (PPE), such as eye and ear protection, gloves, and a dust mask.
- Before starting any work, carefully read and follow the instructions provided by the manufacturer.
- Always use tools and equipment in the way they are intended to be used, and never modify them unless specifically authorized to do so.
- Make sure all equipment is properly grounded and that power cords are in good condition.

- Always keep the work area clean and free of clutter, and ensure that proper ventilation is available.
- Be aware of the location of fire extinguishers, first aid kits, and emergency exits.

### **Engineering Materials:**

- Mild steel is a common engineering material that is relatively inexpensive and has good strength and durability.
- Medium carbon steel is stronger than mild steel and is commonly used for making machine parts and tools.
- High carbon steel has an even higher strength than medium carbon steel and is often used for making cutting tools.
- High speed steel is a type of tool steel that is known for its ability to maintain its hardness even at high temperatures.
- Cast iron is a material that is commonly used for making engine blocks and other heavy-duty parts due to its high strength and durability.

Remember, working in a workshop can be dangerous if proper safety measures are not taken. Always take the time to properly organize your workspace and follow all safety guidelines to ensure a safe and efficient working environment. Additionally, having a good understanding of the different engineering materials that are commonly used in the workshop can help you choose the best material for your specific project.

### **To study and use of different types of tools, equipment, devices & machines used in fitting, sheet metal and welding section.**

In order to excel in the field of fitting, sheet metal, and welding, it is essential to have a good understanding of the tools, equipment, devices, and machines used in this sector. Here are some of the most commonly used tools that you should be familiar with:

1. Hand Tools:
  - Screwdrivers
  - Pliers
  - Wrenches
  - Hammers
  - Chisels
  - Files
  - Saws
  - Snips
  - Clamps
  - Spanners
  - Punches
2. Power Tools:
  - Drills

- Grinders
  - Sanders
  - Shears
  - Planers
  - Routers
  - Welding machines
  - Plasma cutters
  - Oxy-acetylene torches
  - Arc welders
3. Measuring and Marking Tools:
- Tape measures
  - Rulers
  - Squares
  - Protractors
  - Levels
  - Calipers
  - Micrometers
  - Dial indicators
4. Safety Equipment:
- Safety glasses
  - Welding helmets
  - Earplugs
  - Respirators
  - Gloves
  - Safety shoes
  - Hard hats
  - Protective clothing
5. Materials:
- Metals (steel, aluminum, brass, copper)
  - Sheet metal
  - Welding rods
  - Flux
  - Filler metal
  - Abrasives

It is important to know the proper techniques for using each tool, as well as the safety precautions that need to be taken when working with them. Always wear the appropriate safety equipment and follow the manufacturer's instructions carefully.

In addition to understanding the tools, it is also essential to have a good understanding of the various machines used in fitting, sheet metal, and welding. Some of the most commonly used machines include: - Band saws - Shears - Press brakes - Plate rolls - Ironworkers - CNC machines - Laser cutters - Plasma cutters

Each machine has its own unique set of features and capabilities, and it is

important to have a good understanding of how to operate them properly.

In conclusion, mastering the tools, equipment, devices, and machines used in fitting, sheet metal, and welding is essential for success in this field. Regular practice, exploration, and study of these tools and machines will help you to develop the necessary skills and knowledge to excel in this industry.

**Least count determination of Vernier calliper, vernier height gauge, and micrometer (Screw gauge) and taking readings over metallic pieces using these instruments.**

In this section, we will discuss the procedure to determine the least count of Vernier calliper, vernier height gauge, and micrometer (Screw gauge) and taking readings over metallic pieces using these instruments. These instruments are widely used in industries and laboratories to measure the dimensions of objects with high accuracy.

**Vernier Calliper**

The least count of Vernier calliper can be determined by the following steps:

1. Place the jaws of the Vernier calliper on a flat surface and close them gently.
2. Note the main scale reading where the zero on the Vernier scale coincides with a mark on the main scale.
3. Open the jaws of the Vernier calliper and note the distance between the two jaws.
4. Count the number of divisions on the Vernier scale that coincide with the main scale.
5. Calculate the least count using the formula:  $\text{Least count} = (1 \text{ Main scale division}) / (\text{Number of divisions on Vernier scale})$

**Vernier Height Gauge**

The least count of Vernier height gauge can be determined by the following steps:

1. Place the Vernier height gauge on a flat surface and adjust the height using the adjustment knob until the base is level.
2. Note the main scale reading where the zero on the Vernier scale coincides with a mark on the main scale.
3. Slide the Vernier scale down until it touches the surface to be measured.
4. Note the Vernier scale reading where it coincides with a mark on the main scale.

5. Count the number of divisions on the Vernier scale that coincide with the main scale.
6. Calculate the least count using the formula:  $\text{Least count} = (1 \text{ Main scale division}) / (\text{Number of divisions on Vernier scale})$

### **Micrometer (Screw Gauge)**

The least count of micrometer (screw gauge) can be determined by the following steps:

1. Place the micrometer on a flat surface and adjust the thimble using the ratchet until the spindle is closed.
2. Note the main scale reading where the zero on the thimble coincides with a mark on the main scale.
3. Open the spindle of the micrometer and place the object to be measured between the spindle and anvil.
4. Close the spindle using the ratchet until it touches the object.
5. Note the thimble reading where it coincides with a mark on the main scale.
6. Count the number of divisions on the thimble that coincide with the main scale.
7. Calculate the least count using the formula:  $\text{Least count} = (1 \text{ Main scale division}) / (\text{Number of divisions on thimble})$

### **Taking readings over metallic pieces using these instruments**

To take readings over metallic pieces using Vernier calliper, vernier height gauge, and micrometer (screw gauge), the following steps can be followed:

1. Place the metallic piece on a flat surface.
2. Open the jaws of the Vernier calliper or Vernier height gauge or spindle of the micrometer and place it on the metallic piece.
3. Close the jaws or spindle gently until it touches the metallic piece.
4. Note the reading on the main scale and the Vernier scale or thimble.
5. Add the main scale and Vernier scale or thimble readings to get the actual reading.
6. Repeat the above steps for multiple readings and calculate the average reading.

In conclusion, the least count of Vernier calliper, vernier height gauge, and micrometer (screw gauge) can be determined by following the above-mentioned steps. These instruments can be used to measure the dimensions of objects with high accuracy by following the steps to take readings over metallic pieces.

## 2 Machine shop 3 Hrs

In this section, we will discuss the basics of a machine shop, which is a place where various machines are used to shape and cut materials into desired sizes and shapes. The following are some of the key points that you should keep in mind when studying this topic:

- A machine shop is typically equipped with a variety of machines such as lathes, mills, grinders, and drill presses. These machines are used to cut, shape, and drill different types of materials, including metal, wood, and plastic.
- Before using any machine, it is important to understand its basic components and how to operate it safely. This includes knowing how to adjust the cutting speed, feed rate, and depth of cut, as well as how to use safety guards and other safety equipment.
- One of the most common machines in a machine shop is the lathe, which is used to turn cylindrical shapes on a workpiece. To use a lathe, you must first secure the workpiece in a chuck or collet, and then use a cutting tool to remove material from the surface of the workpiece as it rotates.
- Another important machine in a machine shop is the mill, which is used to cut and shape flat surfaces on a workpiece. A mill typically consists of a rotating cutting tool that moves across the surface of the workpiece to remove material and create the desired shape.
- Grinding machines are also commonly found in machine shops, and are used to smooth and shape surfaces on a workpiece. These machines use a rotating abrasive wheel to remove material from the surface of the workpiece, and can be used to create precise shapes and finishes.
- Drill presses are used to drill holes in a workpiece, and are commonly used in metalworking and woodworking applications. To use a drill press, you must first secure the workpiece in place, and then lower the drill bit into the workpiece to create the desired hole.
- In addition to these basic machines, machine shops may also be equipped with specialized machines such as wire EDM machines, waterjet cutters, and CNC machines. These machines are used to create complex shapes and designs, and are typically operated by trained professionals.
- When working in a machine shop, it is important to follow proper safety procedures at all times. This includes wearing proper safety equipment such as eye protection and hearing protection, as well as following established protocols for operating each machine.
- Finally, it is important to maintain and repair machines regularly to ensure that they are operating safely and effectively. This includes cleaning and oiling machines regularly, as well as inspecting them for signs of wear and

damage. If a machine is not functioning properly, it should be repaired or replaced immediately to prevent accidents or injuries.

By keeping these key points in mind, you can gain a better understanding of the basics of a machine shop and the various machines that are used to shape and cut materials. With proper training and practice, you can become a skilled operator of these machines and create high-quality products with precision and accuracy.

### **Demonstration of working, construction and accessories for Lathe machine Perform operations on Lathe - Facing, Plane Turning , step turning, taper turning, threading, knurling and parting.**

Lathe machine is a versatile tool used for shaping and cutting materials such as wood, metal and plastics. It is commonly used in manufacturing and engineering industries to create different shapes and designs. In this article, we will discuss the working, construction and accessories of a lathe machine, as well as the different operations that can be performed on it.

## **Working and Construction of Lathe Machine**

A lathe machine consists of various components, including the bed, headstock, tailstock, tool post, carriage and chuck. The bed is the base of the lathe machine and provides support to all other components. The headstock houses the spindle, which rotates the material being worked on. The tailstock supports the other end of the material and can be moved along the bed to adjust the length of the workpiece. The tool post holds the cutting tool and can be moved along the carriage to adjust the depth and angle of the cut.

## **Accessories for Lathe Machine**

Lathe machine can be equipped with various accessories to enhance its functionality. Some of the commonly used accessories include:

- Chuck: Used to hold the workpiece securely in place.
- Faceplate: Used to hold large and irregularly shaped workpieces.
- Centers: Used to support the material being worked on.
- Collet: Used to hold small and delicate workpieces.
- Taper attachment: Used to perform taper turning operations.

## **Perform Operations on Lathe**

Lathe machine can be used to perform various operations such as:

- Facing: The process of creating a smooth and flat surface on the end of a workpiece.
- Plane turning: The process of removing material from the surface of a workpiece to create a flat surface.

- Step turning: The process of creating multiple steps or levels on the surface of a workpiece.
- Taper turning: The process of creating a tapered surface on the end of a workpiece.
- Threading: The process of creating threads on the surface of a workpiece.
- Knurling: The process of creating a pattern of small raised lines on the surface of a workpiece for better grip.
- Parting: The process of cutting off a section of a workpiece to create a separate piece.

In conclusion, a lathe machine is a versatile tool that can be used to perform various operations on different types of materials. By understanding its working, construction and accessories, one can effectively use a lathe machine to create different shapes and designs as per the requirement.

### **3 Fitting Shop 3 Hrs**

A fitting shop is an essential part of any manufacturing unit where various components are fabricated and assembled to make the final product. In this section, we will discuss the various aspects of a fitting shop and the operations that are carried out there.

#### **1. Tools and Equipment**

- A fitting shop is equipped with a range of tools and equipment that are required for various operations such as drilling, tapping, reaming, shaping, and cutting.
- Some of the common tools found in a fitting shop include hammers, screwdrivers, wrenches, pliers, chisels, saws, and files.
- The equipment found in a fitting shop includes drilling machines, lathes, milling machines, shaping machines, and grinding machines.

#### **2. Operations**

- The fitting shop is responsible for various operations such as marking, cutting, drilling, tapping, reaming, and shaping of various components.
- The marking operation involves marking the components at the required locations to indicate the position where the drilling, cutting, or shaping operations need to be carried out.
- The cutting operation involves cutting the components to the required size and shape using tools such as saws, shears, and cutters.
- The drilling, tapping, and reaming operations involve creating holes in the components using drilling machines and other tools.
- The shaping operation involves shaping the components to the required shape and size using shaping machines and other tools.

#### **3. Safety Measures**

- Safety is a critical aspect of any fitting shop, and various safety measures need to be implemented to ensure the safety of the workers.
- The workers in the fitting shop need to wear appropriate personal protective equipment such as safety goggles, gloves, and aprons to protect themselves from the hazards.
- The machines in the fitting shop need to be regularly maintained and inspected to ensure they are in good working condition.
- The workers need to be trained on the proper use of tools and equipment and the safety measures that need to be taken while working in the fitting shop.

In conclusion, a fitting shop is a crucial part of any manufacturing unit where various components are fabricated and assembled to make the final product. The tools and equipment, operations, and safety measures discussed in this section are essential for the smooth functioning of a fitting shop.

## 1. Practice marking operations.

When it comes to marking up text, there are various operations that you can perform to format your content. Here are some of the most common marking operations that you should practice:

1. **Bold text:** To make text appear bold, surround it with double asterisks (**`**text**`**). This is useful for emphasizing important words or phrases.
2. **Italic text:** To make text appear italicized, surround it with single asterisks (*`*text*`*). This is useful for indicating titles or adding emphasis to certain words.
3. **Underline text:** To underline text, surround it with double underscores (`__text__`). This is useful for indicating headings or important sections.
4. **Strikethrough text:** To add a strikethrough to text, surround it with double tildes (~~`~~text~~`~~). This is useful for indicating deleted or outdated information.
5. **Ordered lists:** To create an ordered list, start each line with a number followed by a period (**1. Item 1, 2. Item 2**, etc.). This is useful for outlining steps or procedures.
6. **Unordered lists:** To create an unordered list, start each line with a hyphen or asterisk (**- Item 1, \* Item 2**, etc.). This is useful for listing items that don't need to be in a specific order.
7. **Headers:** To create a header, start the line with one or more hash symbols (**# Header 1, ## Header 2**, etc.). This is useful for indicating different sections of a document.
8. **Links:** To create a hyperlink, surround the linked text with square brackets and the URL with parentheses (**[text] (url)**). This is useful for pro-

viding additional information or linking to external resources.

9. **Images:** To insert an image, use an exclamation mark followed by square brackets containing the alt text and parentheses containing the image URL (`!alt text`). This is useful for adding visual aids to your content.

By practicing these marking operations, you will be able to format your content in a professional and organized way. It's important to remember that consistency is key, so be sure to use the same formatting throughout your document.

## **2. Preparation of U or V-Shape Male Female Workpiece which contains: Filing, Sawing, Drilling, Grinding.**

When it comes to metalworking, preparing a U or V-shape male-female workpiece is a common task. This workpiece can be used in various applications, including bolts, nuts, and other metal components. In order to prepare this type of workpiece, several steps are involved, including filing, sawing, drilling, and grinding. Here are the steps involved in preparing a U or V-shape male-female workpiece:

1. **Select the Metal:** The first step in preparing a U or V-shape male-female workpiece is to select the metal that will be used. The metal should be strong and durable enough to withstand the intended use of the workpiece. Common metals used for this type of workpiece include steel, aluminum, and brass.
2. **Cut the Metal:** Once the metal has been selected, the next step is to cut it into the desired size and shape. A saw can be used to cut the metal into the desired shape, such as a U or V shape.
3. **File the Edges:** After cutting the metal, the next step is to file the edges to remove any burrs and rough spots. This will ensure that the edges are smooth and will fit together properly.
4. **Drill Holes:** Depending on the intended use of the workpiece, holes may need to be drilled into the metal. A drill can be used to create the necessary holes in the metal.
5. **Grind the Metal:** Finally, the metal can be ground to smooth out any rough spots and ensure a perfect fit between the male and female components of the workpiece.

By following these steps, a U or V-shape male-female workpiece can be prepared for use in a variety of metalworking applications. It is important to take care during each step of the process to ensure that the workpiece is strong, durable, and fits together perfectly.

## 4 Carpentry Shop 3 Hrs

In this section, we will explore the basics of carpentry shop, which is a crucial area in woodworking. The carpentry shop is a place where woodworkers create and assemble wooden structures, furniture, and various wood-based products. Let's dive into the key points of carpentry shop practices:

1. Safety Measures:
  - Always wear protective gear, including goggles, gloves, and earplugs.
  - Keep the shop clean and organized to avoid accidents.
  - Use proper tools for the job and maintain them regularly.
  - Never operate power tools without proper training and supervision.
2. Basic Tools:
  - Hand tools such as hammers, saws, chisels, and planes are essential for carpentry work.
  - Power tools like circular saws, jigsaws, drills, and routers can speed up the work process.
  - Measuring tools such as rulers, squares, and levels are crucial for accuracy.
3. Wood Selection:
  - Choose the right type of wood for the project, considering factors like strength, durability, and appearance.
  - Check for defects such as knots, splits, and warping before purchasing the wood.
  - Consider the wood grain and texture when selecting pieces for a project.
4. Joinery Techniques:
  - Joinery refers to the process of connecting two pieces of wood together.
  - Common joinery techniques include butt joints, lap joints, mortise and tenon joints, and dovetail joints.
  - The choice of joinery technique depends on the project's requirements and the woodworker's skill level.
5. Finishing:
  - Finishing refers to the final steps of a carpentry project, which includes sanding, staining, and sealing the wood.
  - Sanding helps smooth the wood surface and remove any imperfections.
  - Staining can enhance the wood's color and grain pattern, while sealing protects the wood from moisture and other elements.

In conclusion, a carpentry shop is a critical area in woodworking that requires proper safety measures, basic tools, wood selection, joinery techniques, and finishing techniques. By following these essential practices, woodworkers can create beautiful, functional, and long-lasting wooden structures and products.

## Study of Carpentry Tools, Equipment and Different Joints

Carpentry is an essential skill that involves the use of various tools and equipment to create structures and furniture. Understanding the different tools and equipment and knowing how to use them correctly is crucial in achieving quality results. Additionally, having knowledge of different joints used in carpentry is essential in creating strong, durable, and aesthetically pleasing structures. In this study, we will cover the following topics:

1. Carpentry Tools and Equipment:
  - Hand Tools: Hammers, Chisels, Saws, Screwdrivers, Planes, Clamps, etc.
  - Power Tools: Drills, Saws, Routers, Sanders, etc.
  - Measuring and Marking Tools: Tape Measures, Squares, Levels, Marking Gauges, etc.
  - Safety Equipment: Gloves, Goggles, Earplugs, Dust Masks, etc.
2. Carpentry Joints:
  - Cross Half Lap Joint: A joint used to join two pieces of wood at right angles.
    - Steps to create a Cross Half Lap Joint.
  - Half Lap Dovetail Joint: A joint that combines the strength of a Dovetail Joint with the surface area of a Half Lap Joint.
    - Steps to create a Half Lap Dovetail Joint.
  - Mortise and Tenon Joint: A joint used to join two pieces of wood at right angles, with one piece having a slot (mortise) and the other having a protrusion (tenon).
    - Steps to create a Mortise and Tenon Joint.
3. Conclusion:
  - Understanding the different carpentry tools and equipment is essential in achieving quality results.
  - Knowledge of different joints is crucial in creating strong, durable, and aesthetically pleasing structures.
  - Proper safety equipment and precautions should always be used when working with carpentry tools and equipment.

## 5 Welding Shop 6 Hrs

Welding is a vital process that joins two or more metal pieces by melting them together. It is an essential technique used in various industries, including construction, manufacturing, and automotive. The welding process requires the use of specialized equipment and tools, and it is crucial to follow safety guidelines to prevent accidents.

In this article, we will discuss the 5 welding shop practices that you should be aware of when working on welding projects for 6 hours.

1. Safety Procedures
  - Before starting any welding project, always wear personal protective equipment (PPE), including welding helmets, gloves, and safety glasses.

- Inspect the equipment, tools, and work area to ensure they are free from any hazards.
  - Keep a fire extinguisher nearby in case of emergencies.
  - Make sure to have adequate ventilation in the welding shop to avoid inhaling harmful fumes.
2. Welding Equipment
    - Familiarize yourself with the welding equipment, including the welding machine, welding electrodes, and filler metals.
    - Make sure the welding machine is in good condition and set to the appropriate settings for the project.
    - Use the appropriate welding electrodes and filler metals based on the type of metal you are welding.
  3. Welding Techniques
    - Use proper welding techniques, including the correct angle, speed, and distance from the metal being welded.
    - Practice good welding habits, such as maintaining a steady hand, controlling the arc length, and keeping the electrode clean.
    - Use the right welding method, such as MIG, TIG, or Stick welding, based on the type of project.
  4. Welding Maintenance
    - Regularly clean the welding equipment to prevent rust and corrosion.
    - Replace any worn-out parts or accessories to avoid equipment failure during the welding process.
    - Store the welding equipment in a dry and safe place when not in use.
  5. Welding Quality
    - Inspect the welding project after completion to ensure it meets the required standards.
    - Use non-destructive testing methods, such as X-ray or ultrasonic testing, to check the quality of the welds.
    - Make sure to follow the welding code and industry standards to produce high-quality welds.

In conclusion, welding is a crucial process used in various industries, and it is essential to practice safety measures, use the right equipment and techniques, maintain the welding equipment, and produce high-quality welds. Follow these 5 welding shop practices when working on welding projects for 6 hours, and you will have a successful and safe welding experience.

**Introduction to BI Standards and Reading of Welding Drawings.**  
**Practice of Making following Operations: Butt Joint, Lap Joint, TIG Welding, MIG Welding**

Welding is a process of joining two or more metals together by heating them to a high temperature and applying pressure. The process involves the use of various welding techniques, which are widely used in the manufacturing and construction industries. In order to ensure that welding is done correctly and safely, it is important to follow the relevant standards and guidelines. In this study material, we will discuss BI standards and reading of welding drawings, as well as practice making the following operations: butt joint, lap joint, TIG welding, and MIG welding.

### **BI Standards**

BI (British Institute) standards are a set of guidelines that provide recommendations for the welding of different types of metals. These standards are important because they ensure that welding is done safely, efficiently, and according to industry best practices. Some of the key BI standards that are relevant to welding include:

- BS EN ISO 15614-1: Specification and qualification of welding procedures for metallic materials - Welding procedure test - Part 1: Arc and gas welding of steels and arc welding of nickel and nickel alloys
- BS EN ISO 9606-1: Qualification testing of welders - Fusion welding - Part 1: Steels
- BS EN 1090-2: Execution of steel structures and aluminium structures - Part 2: Technical requirements for steel structures

### **Reading Welding Drawings**

Welding drawings are used to communicate important information about the welding process, including the type of weld, the size of the weld, and the location of the weld. In order to read welding drawings effectively, it is important to understand the following:

- Weld symbols: These symbols are used to indicate the type of weld that is required (e.g. fillet weld, butt weld, etc.).
- Dimensions: These indicate the size of the weld, as well as the distance between the welds and other features on the drawing.
- Material requirements: These indicate the type of metal that is being welded, as well as any other materials that are required (e.g. flux, filler wire, etc.).
- Welding process: This indicates the type of welding process that is required (e.g. TIG welding, MIG welding, etc.).

### **Butt Joint**

A butt joint is a type of joint where two pieces of metal are joined together by welding the edges together. This type of joint is commonly used in the manufacturing of pipes, tanks, and other structures. To make a butt joint, the following steps must be followed:

- Prepare the edges of the metal so that they are clean and free from any contaminants.
- Position the two pieces of metal so that the edges are touching.
- Apply heat to the edges of the metal using a welding torch.
- Add filler metal to the joint as required.
- Allow the joint to cool and solidify.

### **Lap Joint**

A lap joint is a type of joint where two pieces of metal are overlapped and welded together. This type of joint is commonly used in the manufacturing of sheet metal products, such as car bodies and airplane wings. To make a lap joint, the following steps must be followed:

- Prepare the edges of the metal so that they are clean and free from any contaminants.
- Position the two pieces of metal so that they are overlapping.
- Apply heat to the edges of the metal using a welding torch.
- Add filler metal to the joint as required.
- Allow the joint to cool and solidify.

### **TIG Welding**

TIG (Tungsten Inert Gas) welding is a type of welding that uses a non-consumable tungsten electrode to produce the weld. This type of welding is commonly used in the manufacturing of high-quality products, such as aircraft parts and medical equipment. To perform TIG welding, the following steps must be followed:

- Prepare the metal so that it is clean and free from any contaminants.
- Position the metal so that it is in the correct location for the weld.
- Begin the arc by striking the tungsten electrode against the metal.
- Add filler metal to the weld as required.
- Control the heat and welding speed to produce a high-quality weld.

### **MIG Welding**

MIG (Metal Inert Gas) welding is a type of welding that uses a consumable wire electrode to produce the weld. This type of welding is commonly used in the manufacturing of automotive parts and other metal products. To perform MIG welding, the following steps must be followed:

- Prepare the metal so that it is clean and free from any contaminants.
- Position the metal so that it is in the correct location for the weld.
- Begin the arc by feeding the wire electrode through the welding torch.
- Control the heat and welding speed to produce a high-quality weld.
- Add shielding gas to protect the weld from contamination.

In conclusion, understanding BI standards, reading welding drawings, and practicing different welding techniques

## **6 Moulding and Casting Shop 6 Hrs**

Moulding and casting are essential techniques in manufacturing, and mastering them can provide an edge in the industry. The following are fundamental concepts and techniques for moulding and casting.

1. Types of Moulds
  - Two-part moulds
  - Waste-moulds
  - Shell moulds
2. Materials for Moulding
  - Plaster of Paris
  - Silicone rubber
  - Polyurethane rubber
  - Latex rubber
3. Techniques for Moulding
  - Pouring the material into the mould
  - Brushing or spraying the material onto the surface
  - Trowelling or spreading the material onto the surface
4. Materials for Casting
  - Resin
  - Metal alloys
  - Plaster of Paris
  - Concrete
5. Techniques for Casting
  - Gravity casting
  - Pressure casting
  - Vacuum casting
6. Post-Casting Techniques
  - Trimming and finishing the casting
  - Polishing the casting
  - Painting or coating the casting

In a six-hour moulding and casting workshop, participants will be able to learn and practice these techniques, as well as create their own moulds and castings. The workshop will include demonstrations, hands-on experience, and guidance from experienced instructors.

By the end of the workshop, participants will have gained a basic understanding of moulding and casting techniques, as well as the ability to create their own moulds and castings. These skills can be applied in various manufacturing industries, such as automotive, aerospace, and consumer goods.

### **Introduction to Patterns**

- Patterns are physical models or templates used to create cavities or shapes in a mould.
- They are made of materials such as wood, metal or plastic.
- Patterns are used in foundry processes to create a mold that can be filled with molten material to create a part.

### **Pattern Allowances**

- Pattern allowances are the dimensional changes made to the pattern to compensate for the contraction of the material during solidification.
- These allowances ensure that the final product has the desired dimensions and surface finish.
- The allowances are based on the material being cast and the specific casting process used.

### **Ingredients of Moulding Sand**

- Moulding sand is a mixture of sand, clay, water and other additives used to create moulds for casting.
- The sand used is typically silica sand, which is high in silica content and has a consistent grain size.
- The clay is used as a binder to hold the sand together and can be bentonite, kaolin or other types of clay.
- Additives such as coal dust or sawdust are added to improve the permeability of the sand.

### **Melting Furnaces**

- Melting furnaces are used to melt the materials used in casting.
- These furnaces can be electric or gas-fired and can reach temperatures of up to 2000°C.
- The type of furnace used depends on the material being melted and the volume of material needed.

### **Foundry Tools and Their Purposes**

- Foundry tools are used to prepare the moulds, create the pattern and remove the casting from the mould.
- Some common tools include:

- Rammers and trowels for compacting the sand in the mould.
- Shovels and scoops for moving sand.
- Crucibles for melting the material.
- Ladles for pouring the molten material into the mould.
- Mallets and chisels for shaping the pattern.

### **Demo of Mould Preparation and Aluminum Casting**

- A demo of mould preparation and aluminum casting can be performed to understand the process firsthand.
- The demonstration will involve creating a pattern, preparing the sand mould, melting the aluminum and pouring it into the mould.
- The finished product will be a small aluminum part.

### **Practice - Study and Preparation of Mould for Plastic**

- To practice the skills learned, a study and preparation of a mould for plastic can be performed.
- This will involve creating a pattern, preparing the mould and melting the plastic to fill the mould.
- The finished product will be a small plastic part.

## **7 CNC Shop 6 Hrs**

CNC (Computer Numerical Control) machines are widely used in the manufacturing industry. It is a highly precise and efficient way to produce complex parts and components. The 7 CNC Shop 6 Hrs is a simulation program that allows individuals to learn and practice operating a CNC machine in a simulated environment. Here are some key points to keep in mind when using the 7 CNC Shop 6 Hrs program:

- The program is designed to simulate a real-world CNC machine, so it is important to approach it with the same level of care and attention as you would if you were using an actual machine.
- The program is divided into different modules that cover various aspects of CNC machining, including programming, tool selection, and operation.
- It is important to start with the basics and work your way up to more advanced modules. This will help you build a solid foundation of knowledge and skills.
- The program includes interactive tutorials that guide you through the various modules. These tutorials provide step-by-step instructions and examples to help you understand the concepts and techniques involved in CNC machining.
- In addition to the tutorials, the program also includes practice exercises that allow you to apply what you have learned in a simulated environment. These exercises help you build confidence and improve your skills.

- The program also includes a feedback system that provides helpful tips and suggestions to help you improve your performance. This feedback is based on your actions in the program and helps you identify areas where you may need to improve.
- It is important to take your time and not rush through the program. CNC machining requires precision and attention to detail, so it is important to approach it with patience and focus.
- Finally, it is important to remember that the 7 CNC Shop 6 Hrs program is a tool to help you learn and practice CNC machining. It is not a substitute for real-world experience and training, but it can be a valuable supplement to your learning and development.

**Study of main features and working parts of CNC machine and accessories that can be used. Perform different operations on metal components using any CNC machines.**

In this study material, we will explore the main features and working parts of a CNC machine and its accessories. Additionally, we will learn how to perform various operations on metal components using any CNC machine.

#### **Features of a CNC Machine**

1. Precision: CNC machines are known for their precise and accurate results. They can produce parts with tolerances as small as a few microns.
2. Automation: CNC machines are fully automated and do not require manual intervention during the machining process. Once the program is loaded, the machine can run for hours without any human input.
3. Flexibility: CNC machines can be used to produce a wide range of parts, from simple to complex geometries. They can also be used to machine a variety of materials, including metals, plastics, and composites.
4. Reproducibility: CNC machines can produce identical parts with the same level of precision and accuracy every time, making them ideal for high-volume production.

#### **Working Parts of a CNC Machine**

1. Control Unit: The control unit is the brain of the CNC machine. It reads the program and sends commands to the other parts of the machine to perform the machining operation.
2. Motors: CNC machines use motors to move the cutting tool and the workpiece. These motors are controlled by the control unit and can move the tool and the workpiece in multiple axes.
3. Spindle: The spindle is the part of the machine that holds the cutting tool. It rotates at high speeds to cut the material.

4. Tool Changer: CNC machines can have automatic tool changers that allow for multiple tools to be used during a single machining operation.

#### **Accessories for CNC Machines**

1. Workholding Devices: Workholding devices are used to hold the workpiece in place during machining. Examples include vices, clamps, and fixtures.
2. Cutting Tools: Cutting tools are used to remove material from the workpiece. They include drills, end mills, and inserts.
3. Toolholders: Toolholders are used to hold the cutting tools in place. They come in various sizes and styles, depending on the type of tool being used.
4. Coolant System: A coolant system is used to cool the cutting tool and the workpiece during machining. This helps to prevent overheating and prolongs the life of the cutting tool.

#### **Operations that can be performed on CNC Machines**

1. Drilling: CNC machines can be used to drill holes in a variety of materials. The size and depth of the holes can be controlled with high precision.
2. Milling: Milling is a common operation performed on CNC machines. It involves removing material from the surface of the workpiece using a rotating cutting tool.
3. Turning: Turning is another common operation performed on CNC machines. It involves rotating the workpiece while a cutting tool removes material from the surface.
4. Grinding: CNC machines can also be used for grinding operations, which involve removing material from the surface of the workpiece using an abrasive wheel.

In conclusion, CNC machines are highly precise, flexible, and automated machines that can produce a wide range of parts with high accuracy and reproducibility. Understanding the main features and working parts of these machines, as well as the accessories and operations that can be performed, is essential for anyone looking to use these machines for metal component manufacturing.

### **8 To prepare a product using 3D printing 3 Hrs**

In order to prepare a product using 3D printing, one must follow a set of steps to ensure a successful outcome. Below are the steps involved in preparing a product using 3D printing within 3 hours:

1. Choose a suitable 3D modeling software: In order to prepare a 3D print, you need to have a 3D model of the product. There are several 3D modeling software options available, such as Tinkercad, Fusion 360, and

SketchUp. Choose a software you are familiar with and begin modeling the product.

2. Design the product: Once you have chosen your 3D modeling software, begin designing the product you want to print. This may involve creating a new model from scratch or modifying an existing one. Make sure the design is suitable for 3D printing and will not cause any issues during the printing process.
3. Export the 3D model: After you have finished designing the product, export the 3D model as an STL file. This is the file format required by most 3D printers.
4. Prepare the 3D printer: Once you have the STL file, set up your 3D printer. This may involve leveling the printer bed, loading the filament, and selecting the appropriate print settings.
5. Slice the 3D model: The next step is to slice the 3D model using slicing software. This software will take the 3D model and convert it into a series of 2D layers that the 3D printer can print. Make sure to select the appropriate settings, such as layer height and infill density.
6. Start the 3D print: Once the slicing is complete, start the 3D print. This may take several hours, depending on the size and complexity of the product. Make sure to monitor the print to ensure it is progressing as expected.
7. Post-process the 3D print: After the print is complete, remove it from the printer and post-process it as needed. This may involve removing any support material or sanding down rough edges.
8. Finish the product: Finally, finish the product as desired. This may involve painting it or adding additional components to complete the product.

By following these steps, you can successfully prepare a product using 3D printing within 3 hours. Make sure to carefully follow each step to ensure a successful outcome.

### **Total 33 Hrs**

In this study material, we will cover the topic of Total 33 Hrs. This topic is important for understanding the concept of time management and how to effectively use the available time to achieve our goals. Below are the key points that will be covered in this study material:

1. Understanding the concept of Total 33 Hrs:
  - Total 33 Hrs refers to the number of hours available in a day for an individual to work on their personal and professional goals.
  - This includes the 24 hours in a day plus the additional 9 hours that can be utilized by effective time management.
2. Importance of effective time management:

- Effective time management can help individuals achieve their goals in a timely and efficient manner.
  - It also helps in reducing stress and increasing productivity.
3. Tips for effective time management:
    - Prioritize tasks based on their importance and urgency.
    - Break down larger tasks into smaller, manageable ones.
    - Set realistic deadlines and stick to them.
    - Avoid procrastination and distractions.
    - Take breaks to avoid burnout and maintain focus.
  4. Benefits of effective time management:
    - Increased productivity and efficiency.
    - Better work-life balance.
    - Reduced stress and anxiety.
    - Improved decision-making skills.
    - More time for personal growth and development.
  5. Tools for effective time management:
    - Time tracking apps such as Toggl and RescueTime.
    - Project management tools such as Asana and Trello.
    - Calendar apps such as Google Calendar and Outlook.
    - Productivity apps such as Focus@Will and Forest.
  6. Conclusion:
    - Total 33 Hrs is an important concept for understanding the value of time management.
    - By effectively utilizing the available time, individuals can achieve their goals in a timely and efficient manner.
    - By following the tips and using the tools mentioned in this study material, individuals can improve their time management skills and achieve success in their personal and professional lives.

### **Course Outcome:**

Upon completion of this course, students will be able to:

- Understand the fundamental concepts and principles of the subject matter.
- Analyze and solve problems related to the subject matter using appropriate tools and techniques.
- Apply the acquired knowledge and skills to real-world scenarios.
- Demonstrate effective communication skills through oral and written presentations.
- Work collaboratively in teams to achieve common goals.
- Develop a strong work ethic and professionalism in their approach to the subject matter.
- Recognize the importance of life-long learning and continuous professional development.
- Evaluate and critique research and literature related to the subject matter.
- Develop a critical and analytical mindset towards the subject matter.

- Demonstrate ethical and responsible behavior in their professional and personal lives.
- Understand the impact of the subject matter on society and the environment.
- Appreciate the diversity and multiculturalism of the subject matter and its relevance in a global context.
- Identify and apply relevant laws, regulations, and ethical standards related to the subject matter.
- Develop a strong foundation for further studies and professional development in the field.

## Introduction

Engineering is a vast field that involves the application of scientific and mathematical principles to design and develop systems, structures, and machines. To become an engineer, it is essential to have a solid understanding of various engineering materials, tools, machines, and measuring equipment. In this study material, we will cover the following topics:

- Blooms Level CO1: Use various engineering materials, tools, machines, and measuring equipment.
- K3 CO2: Perform machine operations in lathe and CNC machine.
- K3 CO3: Perform manufacturing operations on components in fitting and carpentry shop.
- K3 CO4: Perform operations in welding, molding, casting, and gas cutting.
- K3 CO5: Fabricate a job by 3D printing manufacturing technique.

### **Blooms Level CO1: Use various engineering materials, tools, machines, and measuring equipment**

To become a successful engineer, it is important to have a good understanding of the different engineering materials, tools, machines, and measuring equipment. Here are some key points to help you understand these concepts:

- Engineering materials: Engineering materials include metals, polymers, ceramics, and composites. They are used to create structures, machines, and other engineering products. It is essential to understand the properties and characteristics of these materials to select the appropriate material for a specific application.
- Tools: Various tools are used in the engineering field, such as hammers, screwdrivers, pliers, wrenches, and many others. These tools are used to assemble and disassemble components, and it is essential to know how to use them correctly.
- Machines: Machines are used to transform the engineering materials into useful products. The lathe and CNC machines are used to perform operations like drilling, turning, and milling.

- Measuring equipment: Measuring equipment is used to measure the dimensions of engineering components. Common measuring equipment includes Vernier calipers, micrometers, and height gauges.

### **K3 CO2: Perform machine operations in lathe and CNC machine**

The lathe and CNC machines are essential tools in the engineering field. Here are some key points to help you understand these concepts:

- Lathe machine: A lathe machine is used to perform operations like drilling, turning, and milling. It is essential to understand the different parts of the lathe machine and how to operate it safely.
- CNC machine: A CNC machine is used to perform complex operations like cutting, drilling, and shaping. It is essential to know how to program the CNC machine and operate it safely.

### **K3 CO3: Perform manufacturing operations on components in fitting and carpentry shop**

Fitting and carpentry shops are essential parts of the engineering field. Here are some key points to help you understand these concepts:

- Fitting shop: A fitting shop is used to manufacture and assemble components using various tools like hand tools, power tools, and welding equipment. It is essential to know how to use these tools safely and correctly.
- Carpentry shop: A carpentry shop is used to manufacture wooden components using various tools like saws, chisels, and planes. It is essential to know how to use these tools safely and correctly.

### **K3 CO4: Perform operations in welding, molding, casting, and gas cutting**

Welding, molding, casting, and gas cutting are essential operations in the engineering field. Here are some key points to help you understand these concepts:

- Welding: Welding is used to join two or more metal components. It is essential to know how to perform welding safely and correctly.
- Molding: Molding is used to manufacture components using molds. It is essential to know how to prepare the mold and the molding material correctly.
- Casting: Casting is used to manufacture components using a mold and molten metal. It is essential to know how to prepare the mold and perform casting safely and correctly.
- Gas cutting: Gas cutting is used to cut metal components using a gas flame. It is essential to know how to use the gas cutting equipment safely and correctly.

### **K3 CO5: Fabricate a job by 3D printing manufacturing technique**

3D printing is a relatively new technology in the engineering field. Here are some key points to help you understand this concept:

- 3D printing: 3D printing is a manufacturing technique that uses a computer to create a three-dimensional object from a digital model. It is essential to know how to prepare the digital model and operate the 3D printer safely and correctly.

### **Conclusion**

In conclusion, engineering is a complex field that requires a solid understanding of various materials, tools, machines, and manufacturing techniques. We hope this study material has helped you understand these concepts better and prepare for your exams. Remember to always prioritize safety and follow the correct procedures when working with engineering materials, tools, machines, and measuring equipment.

### **Reference Books:**

When it comes to studying for exams or enhancing your knowledge in a particular subject, reference books can be incredibly helpful. Here are some tips on choosing and using reference books effectively:

1. Look for books that are written by experts in the field. These authors often have years of experience and can provide valuable insights and information.
2. Choose books that are up-to-date. Information in certain fields can change rapidly, so it's important to have the most current information available.
3. Consider the level of detail you need. Some reference books provide a broad overview of a subject, while others go into great detail. Choose the book that best suits your needs.
4. Use the table of contents and index to your advantage. These tools can help you find the information you need quickly and efficiently.
5. Take notes as you read. This can help you remember key information and also make it easier to review later.
6. Don't be afraid to ask for help. If you're struggling to understand a concept or need clarification, reach out to a teacher, tutor, or colleague for assistance.
7. Keep your reference books organized and easily accessible. This will save you time and frustration when you need to find information quickly.

Remember, reference books are a valuable tool for learning and expanding your knowledge. By choosing the right books and using them effectively, you can

achieve great success in your studies and career.

### **1. Workshop Practice, H S Bawa, McGraw Hill**

Workshop Practice is an essential aspect of engineering studies, and it involves the application of theoretical knowledge in practical situations. The book “Workshop Practice” by H S Bawa, published by McGraw Hill, is a comprehensive guide for students to understand the various aspects of workshop practice. Here are some key points covered in the book:

1. Introduction to workshop practice: The book provides an overview of the workshop environment, safety measures, and the tools and equipment used in a workshop.
2. Workshop tools: The book covers the different types of hand tools such as hammers, pliers, screwdrivers, wrenches, and spanners, along with their uses.
3. Measuring tools: The book explains the various types of measuring tools used in a workshop, including micrometers, vernier calipers, dial indicators, and height gauges.
4. Cutting tools: The book covers the different types of cutting tools such as files, saws, chisels, and drills, along with their uses and maintenance.
5. Fastening and joining: The book explains the different types of fasteners such as nuts, bolts, screws, and rivets, along with their uses and applications.
6. Welding and brazing: The book covers the different types of welding and brazing processes used in a workshop, including gas welding, arc welding, and TIG welding.
7. Machining processes: The book provides an overview of the different types of machining processes such as turning, milling, drilling, and grinding, along with their applications.
8. Sheet metal working: The book covers the different types of sheet metal working processes such as cutting, bending, and forming, along with their applications.
9. Casting and molding: The book explains the different types of casting and molding processes used in a workshop, including sand casting, die casting, and injection molding.
10. Maintenance and safety: The book emphasizes the importance of maintaining the tools and equipment used in a workshop and provides guidelines for ensuring safety in the workshop environment.

In conclusion, “Workshop Practice” by H S Bawa is a valuable resource for students studying engineering and related fields. The book covers a wide range

of topics related to workshop practice and provides practical knowledge and guidance for students to apply in their future careers.

## **2. Mechanical Workshop Practice, K C John, PHI**

Mechanical Workshop Practice is an important subject in the field of mechanical engineering. It is a practical course that teaches students about the various tools, machines, and processes used in mechanical workshops. This subject is essential for students who want to work in manufacturing industries or pursue a career in mechanical engineering.

The following are the key topics covered in Mechanical Workshop Practice:

1. Introduction to Workshop Practice
  - Definition of Workshop Practice
  - Types of workshops
  - Safety precautions in the workshop
  - Basic hand tools used in the workshop
2. Workshop Processes
  - Marking and layout tools
  - Cutting tools
  - Grinding and finishing tools
  - Measuring tools
  - Welding and brazing
  - Casting and forging
  - Sheet metal work
3. Machine Tools
  - Types of machine tools
  - Lathe machine
  - Drilling machine
  - Shaping machine
  - Milling machine
  - Grinding machine
4. Mechanical Measurements
  - Use of Vernier caliper, micrometer, and other measuring instruments
  - Measurement of length, diameter, and thickness
  - Measurement of angles and surface finish
5. Computer-Aided Manufacturing (CAM)
  - Introduction to CAM
  - Types of CAM systems
  - Advantages and disadvantages of CAM
  - Applications of CAM in manufacturing

## 6. Quality Control

- Definition of quality control
- Types of quality control
- Statistical quality control
- Quality control tools and techniques

In conclusion, Mechanical Workshop Practice is a crucial subject for mechanical engineering students. It provides hands-on training in the use of tools, machines, and processes used in mechanical workshops. By studying this subject, students can gain practical skills and knowledge that can help them excel in the field of mechanical engineering.

### **3. Workshop Practice Vol 1, and Vol 2, by HazraChoudhary , Media promoters and Publications**

Workshop Practice Vol 1, and Vol 2 are two books written by HazraChoudhary and published by Media promoters and Publications. These books are helpful for students who want to learn about workshop practices and techniques. Below are some key points about these books:

#### **Workshop Practice Vol 1**

1. This book introduces the reader to the basic concepts of workshop practices and provides a comprehensive overview of various tools and equipment used in a workshop.
2. It covers topics such as marking out, drilling, tapping, reaming, and grinding, among others.
3. The book also includes detailed information on the use of hand tools, power tools, and measuring instruments.
4. It provides step-by-step instructions and illustrations for various workshop operations, making it easy for students to understand and learn.
5. The book also includes a section on safety precautions to be taken while working in a workshop, which is important for students to understand before starting any practical work.

#### **Workshop Practice Vol 2**

1. This book focuses on advanced workshop practices and techniques and is a continuation of Workshop Practice Vol 1.
2. It covers topics such as lathe operations, milling operations, shaping operations, and grinding operations.
3. The book also includes a section on welding and soldering techniques, which are important for students who want to pursue a career in manufacturing or engineering.
4. It provides detailed information on the use of different types of machines and tools used in advanced workshop practices.

5. The book also includes a section on quality control and inspection, which is important for students who want to ensure that their work meets industry standards.

In conclusion, Workshop Practice Vol 1, and Vol 2, by HazraChoudhary, are two important books for students who want to learn about workshop practices and techniques. These books provide comprehensive coverage of various tools and equipment used in a workshop, step-by-step instructions and illustrations for various workshop operations, and safety precautions to be taken while working in a workshop. Students who read these books will gain a better understanding of workshop practices and will be better prepared for hands-on practical work.

#### **4. CNC Fundamentals and Programming, By P. M. Agrawal, V. J. Patel, Charotar Publication.**

CNC or Computer Numerical Control is a technology that is used to control machines using computer software. The CNC technology is used in a variety of industries such as aerospace, automotive, and manufacturing. The CNC technology has revolutionized the manufacturing industry by making it possible to produce complex parts with high precision and accuracy.

The CNC Fundamentals and Programming book by P. M. Agrawal and V. J. Patel, published by Charotar Publication, provides a comprehensive guide to CNC technology and programming. Following are the important topics covered in this book:

1. Introduction to CNC Technology: The book provides an introduction to CNC technology, including its history, advantages, and applications in different industries.
2. CNC Machine Components: The book covers the different components of a CNC machine, including the controller, motors, drives, and feedback devices.
3. CNC Programming Fundamentals: The book covers the basics of CNC programming, including G-codes, M-codes, and other programming commands.
4. CNC Programming Techniques: The book provides various programming techniques used in CNC programming, such as canned cycles, subroutines, and parametric programming.
5. CNC Programming for Turning Operations: The book covers the programming techniques used for turning operations, including roughing, finishing, and threading.
6. CNC Programming for Milling Operations: The book covers the programming techniques used for milling operations, including contour milling, pocket milling, and drilling.

7. CNC Programming for Drilling Operations: The book covers the programming techniques used for drilling operations, including peck drilling, deep-hole drilling, and drilling cycles.
8. CNC Programming for Grinding Operations: The book covers the programming techniques used for grinding operations, including surface grinding, cylindrical grinding, and centerless grinding.
9. CNC Programming for EDM Operations: The book covers the programming techniques used for EDM operations, including wire EDM and sinker EDM.
10. CNC Programming for Laser Cutting Operations: The book covers the programming techniques used for laser cutting operations, including contour cutting and piercing.

In conclusion, the CNC Fundamentals and Programming book by P. M. Agrawal and V. J. Patel is an excellent resource for anyone interested in learning about CNC technology and programming. The book provides a comprehensive guide to CNC technology and programming, covering all the important topics in a clear and concise manner.